A FINITE EULER–CLAUSEN TRANSFER FOR APÉRY-LIKE SUMS, WITH AN APPLICATION TO A CONJECTURE OF ZHI-HONG SUN

HUIMIN ZHENG

ABSTRACT. Let $G_n = \sum_{k=0}^{n} \binom{2k}{k}^2 \binom{2n-2k}{n-k} 4^{n-k}$ be the Apéry-like sequence of Almkvist–Zudilin. We study the finite sums $\sum_{n<p} \binom{2n}{n} G_n/M^n$ and their first moments $\sum_{n<p} n\binom{2n}{n} G_n/M^n$ modulo powers of an odd prime p. The generating series $H(z) = \sum_{n\geq 0} \binom{2n}{n} G_n z^n/64^n$ satisfies the exact identity $H(z) = K(4z(1-z))$, where $K(w) = {}_3F_2(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}; 1, 1; w)$. We prove that truncating this identity at degree p commutes with specialization at the center $z = \frac{1}{2}$ up to p^3, and that its Euler derivative at any p-integral point with $1 - z$ a p-unit is correct up to p^2; in both cases the first defect is computed explicitly and shown to possess a reflection symmetry, a constant-coefficient forcing equation and a short hypergeometric kernel. Combining these finite transfers with published special values of truncated triple-binomial sums (Sun, Sun–Ye) and with an explicit specialization of the general CM theorem of Chisholm–Deines–Long–Nebe–Swisher, we prove all nine formulas of Conjecture 2.4 of Z.-H. Sun [*New congruences involving Apéry-like numbers*, arXiv:2004.07172v2]: three pairs of weighted congruences modulo p^2 with denominators $(72, 576)$, $(48, -192)$, $(63, -4032)$, and the unweighted congruence with denominator 128 modulo p^3. Selected algebraic components of the argument have been checked in Lean.

1. INTRODUCTION

For a nonnegative integer n, the Apéry-like sequence studied in this paper is

$$G_n = \sum_{k=0}^{n} \binom{2k}{k}^2 \binom{2n-2k}{n-k} 4^{n-k} \tag{1}$$

(OEIS A143583; $1, 12, 164, 2352, 34596, \dots$). It is one of the sporadic second-kind Apéry-like sequences of Almkvist–Zudilin [1] and satisfies

$$(n+1)^2 G_{n+1} = (32n(n+1) + 12)G_n - 256n^2 G_{n-1} \qquad (n \geq 1).$$

2020 *Mathematics Subject Classification.* Primary 11A07; Secondary 11B65, 11E25, 33C20, 11F33.

Key words and phrases. Apéry-like numbers; supercongruence; truncated hypergeometric series; Clausen's identity; Ramanujan-type series for $1/\pi$; binary quadratic forms.

ORCID: 0009-0004-8605-2713.

Zhi-Hong Sun [12] initiated a systematic study of the p-adic behavior of the sums

$$\sum_{n=0}^{p-1}\frac{\binom{2n}{n}G_n}{M^n},\qquad \sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{M^n}, \tag{2}$$

and stated there a number of precise conjectures. The present paper is concerned with the following one [12, Conjecture 2.4].

Conjecture (Z.-H. Sun). *Let $p>3$ be a prime. Then*

$$\begin{aligned}
\sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{72^n}&\equiv -8\sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{576^n}\\
&\equiv\begin{cases}4x^2-3p \pmod{p^2} & \text{if } p=x^2+4y^2\equiv 1 \pmod 4,\\ p \pmod{p^2} & \text{if } p\equiv 3 \pmod 4,\end{cases}\\
\sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{48^n}&\equiv 4\sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{(-192)^n}\\
&\equiv\begin{cases}3p-4x^2 \pmod{p^2} & \text{if } p=x^2+3y^2\equiv 1 \pmod 3,\\ -p \pmod{p^2} & \text{if } p\equiv 2 \pmod 3,\end{cases}\\
\sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{63^n}&\equiv 64\sum_{n=0}^{p-1}\frac{n\binom{2n}{n}G_n}{(-4032)^n}\\
&\equiv\begin{cases}-8(4x^2-3p) \pmod{p^2} & \text{if } p=x^2+7y^2\equiv 1,2,4 \pmod 7,\\ -8p \pmod{p^2} & \text{if } p\equiv 3,5,6 \pmod 7,\end{cases}
\end{aligned}$$

and

$$\sum_{n=0}^{p-1}\frac{\binom{2n}{n}G_n}{128^n}\equiv\begin{cases}4x^2-2p-\dfrac{p^2}{4x^2} \pmod{p^3} & \text{if } p=x^2+2y^2\equiv 1,3 \pmod 8,\\ \dfrac{p^2}{3}\binom{\lfloor p/4\rfloor}{\lfloor p/8\rfloor}^{-2} \pmod{p^3} & \text{if } p\equiv 5 \pmod 8,\\ -\dfrac{3p^2}{2}\binom{\lfloor p/4\rfloor}{\lfloor p/8\rfloor}^{-2} \pmod{p^3} & \text{if } p\equiv 7 \pmod 8.\end{cases}$$

(The third pair involves the denominator 63, so at $p=7$ the sums are not p-integral and the statement is understood for $p\neq 7$.)

Why these sums are natural, and what was missing. Sums of the shape (2) are the Apéry-side incarnation of a classical theme going back to Van Hamme [17] and Rodriguez-Villegas [10]: truncated hypergeometric series at CM points are expected to be congruent, modulo p^2 or p^3, to quantities built from binary quadratic forms. For the *triple binomial sums*

$$T_p(d)=\sum_{k=0}^{p-1}\frac{a_k}{d^k},\qquad U_p(d)=\sum_{k=0}^{p-1}\frac{k\,a_k}{d^k},\qquad a_k=\binom{2k}{k}^2\binom{4k}{2k}=\frac{(4k)!}{(k!)^4}, \tag{3}$$

an extensive theory now exists: Sun evaluated $U_p(648)$ and $U_p(-144)$ modulo p^2 [13, Theorems 2.9, 2.10], Sun–Ye evaluated $T_p(d)$ modulo p^3 at several CM points including $d=256$ and $d=-3969$ [15, Theorems 1.5, 1.8], and

a general theorem of Chisholm–Deines–Long–Nebe–Swisher [4, Theorem 1] evaluates truncated CM Ramanujan-type sums modulo p^2, covering both ordinary and supersingular primes.

The sums (2), however, involve the Apéry-like numbers G_n themselves, not the triple binomial coefficients a_k. What separates the two is a *finite transformation problem*: the generating series of $\binom{2n}{n}G_n/64^n$ is related to the ${}_3F_2$ series $K(w)$ by the quadratic substitution $w = 4z(1-z)$ (Theorem 2.2 below), but *truncating* an identity of formal power series at degree p and then specializing at a point is not congruence-compatible in general: the discarded tail $k > (p-1)/2$ has coefficients divisible only by p^2 (Lemma 2.1), and at a generic point this residual p^2 is exactly what one sees. The content of the present paper is that at the center $z = \frac{1}{2}$ a further factor p appears (so the center comparison holds modulo p^3), while the first p^2-defect at a generic point is an explicit polynomial $R_p(z)$ whose *odd reflection symmetry* is responsible for the pairing relations in Sun's conjecture. Controlling this defect — not evaluating any new CM period — is the technical heart of the matter.

Main results. Throughout, p is an odd prime, $\mathbb{Z}_{(p)}$ denotes the localization of $\mathbb{Z}$ at p, and

$$H_p(z) = \sum_{n=0}^{p-1} \frac{\binom{2n}{n}G_n}{64^n} z^n, \qquad K_p(w) = \sum_{k=0}^{p-1} \frac{a_k}{256^k} w^k.$$

- **Theorem 2.2** (formal identity). $H(z) = K(4z(1-z))$ in $\mathbb{Q}[[z]]$; equivalently $H(z) = B(z)^2$ with $B(z) = {}_2F_1(\frac{1}{4}, \frac{3}{4}; 1; z)$.
- **Theorem 3.1** (center bridge). For *every* odd prime p,
$$\sum_{n=0}^{p-1} \frac{\binom{2n}{n}G_n}{128^n} \equiv \sum_{k=0}^{p-1} \frac{a_k}{256^k} \pmod{p^3}.$$
- **Theorem 3.3** (first defect at the center). The quotient of the difference by p^3 is given modulo p by an explicit finite double sum (12); in particular the center comparison does *not* lift to p^4 in general.
- **Theorem 4.1** (Euler transfer). $H_p(z) - K_p(4z(1-z)) \in p^2\mathbb{Z}_{(p)}[z]$, and the first defect has the form $p^2 z^p R_p(z) \pmod{p^3}$ with $R_p(1-z) = -R_p(z)$.
- **Corollary 4.3** (pairing and one-sided evaluations). For every z with $z, 1-z \in \mathbb{Z}_{(p)}^{\times}$,
$$(1-z)\,\mathcal{W}_p(z) + z\,\mathcal{W}_p(1-z) \equiv 0 \pmod{p^2},$$
$$\mathcal{W}_p(z) \equiv \frac{1-2z}{1-z}\,(\theta_w K_p)(w(z)) \pmod{p^2},$$
where $\mathcal{W}_p(z) = zH_p'(z)$ and $\theta_w = w\,\mathrm{d}/\mathrm{d}w$.
- **Theorem 5.1** (structure of the defect). $\mathcal{S}R_p = \varepsilon/4$ in $\mathbb{F}_p[z]$ for an explicit third-order operator $\mathcal{S}$ and $\varepsilon = \left(\frac{-1}{p}\right)$; R_p factors as

$(1-2z)J_p(4z(1-z))$ with an explicit hypergeometric kernel J_p of degree $\lfloor (p-3)/4 \rfloor$; and R_p is determined by a polynomial integral invariant with a characteristic-p correction term z^p.

- **Theorems 7.1 and 7.4** (denominator 128 modulo p^3). The last displayed formula of the conjecture holds: the split branch by combining Theorem 3.1 with [15, Theorem 1.5], the inert branches by combining it with a jet comparison (Proposition 7.3) and an exact rational identity $\mathcal{Q}_a(\frac{1}{2}) = -\big(2^{a-1}/(a\binom{a-1}{(a-1)/2})\big)^2$ for odd a (Theorem 7.2) proved by a symmetric-square method.
- **Theorem 8.1**. All nine formulas of Sun's Conjecture 2.4 hold.

What is new and what is cited. The finite transfers (Theorems 3.1, 3.3, 4.1, 5.1), the symmetric-square identity (Theorem 7.2) and the assembly are proved here. All CM special values of the triple binomial sums are taken from the literature: [13, Theorems 2.9, 2.10 and Remark 2.3], [15, Theorems 1.5, 1.8], [11, Theorem 5.1], and the general CM theorem [4, Theorem 1], whose specialization to the $65k+8$ series at $\lambda = -256/3969$ we verify in detail, including the two primes $p = 5, 13$ excluded by its unit hypotheses (Proposition 6.3). The classical identity $\sum_{k\geq 0}(65k+8)a_k/(-3969)^k = 9\sqrt{7}/\pi$ [5, Table 2] is used only to identify the parameters of that theorem. Mao–Liu [7, Theorem 1.10] have obtained the *quadratic* and $1/(2n-1)$ moments of the denominator-128 sum modulo p^2 by a different reduction; their results neither contain nor follow from the present ones (different weights, different modulus). The unweighted denominator-128 sum modulo p^3 and the linear moments at $72, 576, 48, -192, 63, -4032$ do not appear in the literature we surveyed; we make no claim of priority beyond this.

Other recent results on the same Apéry-like family concern parameter moments without the additional central-binomial weight [8, 6], or Christoffel–Darboux identities and endpoint congruences for the polynomial families $g_n(x)$ and $v_n(x)$ [14]. These works provide useful context but do not furnish the finite quadratic pullback used here. This distinction is important: the new step in the present argument is not another isolated CM evaluation, but control of the error created by truncating that pullback.

Outline. Section 2 proves the formal identity. Section 3 proves the center bridge and its first defect. Section 4 proves the Euler transfer and the pairing corollary. Section 5 analyses the defect polynomial R_p. Section 6 assembles the three weighted pairs from the cited special values, with the exceptional primes settled directly. Section 7 proves the denominator-128 formulas modulo p^3. Section 8 states the final corollary and some remarks, including an obstruction to a naive finite Pfaff transformation. Appendix A collects the finite exceptional-prime certificates; Appendices B and C give the self-contained proofs of the jet comparison and of the symmetric-square identity; Appendix D describes the computational checks.

Acknowledgments. The proofs and the initial manuscript draft were developed with substantial assistance from OpenAI's Codex. The author reviewed the mathematical arguments, computational checks, references, and final presentation and assumes responsibility for the contents of the article.

Data availability. No datasets were generated or analysed in this work. The finite computations described in Appendix D are reproducible from the displayed definitions.

Competing interests. The author declares no competing interests.

2. Notation and the generating-function identity

Throughout p is an odd prime and v_p the p-adic valuation. We set

$$a_k = \binom{2k}{k}^2 \binom{4k}{2k} = \frac{(4k)!}{(k!)^4}, \qquad c_k = \frac{a_k}{256^k}, \qquad h_n = \frac{\binom{2n}{n} G_n}{64^n},$$

$$H(z) = \sum_{n\geq 0} h_n z^n, \qquad K(w) = \sum_{k\geq 0} c_k w^k = {}_3F_2\Big(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{3}{4}; 1, 1; w\Big),$$

and denote by H_p, K_p the truncations at degree $p-1$. Write

$$s = z(1-z), \qquad t = 1-2z, \qquad w = w(z) = 4s.$$

Lemma 2.1 (valuation)**.** *For $0 \leq k < p$ one has $v_p(a_k) = v_p(c_k) = \lfloor 4k/p \rfloor$; in particular $v_p(c_k) \geq 2$ for $k > (p-1)/2$, and $v_p(c_k) \geq 3$ for $k \geq (3p+1)/4$.*

Proof. Since $k < p$, $v_p(k!) = 0$, and $4k \leq 4(p-1) < p^2$ because $p^2 - 4(p-1) = (p-2)^2 > 0$; hence $v_p(a_k) = v_p((4k)!) = \lfloor 4k/p \rfloor$. □

Theorem 2.2 (formal identity)**.** *In $\mathbb{Q}[[z]]$,*

$$H(z) = K(4z(1-z)).$$

Equivalently $H(z) = B(z)^2$, where

$$B(z) = {}_2F_1\Big(\frac{1}{4}, \frac{3}{4}; 1; z\Big) = \sum_{n\geq 0} \binom{2n}{n}\binom{4n}{2n} \frac{z^n}{64^n}.$$

Proof. Step 1: a second-order equation for $g(x) = \sum_{n\geq 0} G_n x^n$. Since $(1-16x)^{-1/2} = \sum_{m\geq 0} \binom{2m}{m}(4x)^m$ and ${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; 16x) = \sum_{k\geq 0} \binom{2k}{k}^2 x^k$, the defining convolution (1) is exactly the Cauchy product

$$g(x) = (1-16x)^{-1/2}\, {}_2F_1\Big(\tfrac{1}{2}, \tfrac{1}{2}; 1; 16x\Big). \tag{4}$$

The hypergeometric factor satisfies the Gauss equation $x(1-16x)u'' + (1-32x)u' - 4u = 0$; substituting $u = (1-16x)^{1/2} g$ and clearing the invertible factor gives

$$x(1-16x)^2 g'' + (1-48x)(1-16x)g' + (256x-12)g = 0,$$

i.e. coefficientwise

$$(n+1)^2 G_{n+1} = (32n(n+1)+12)G_n - 256n^2 G_{n-1} \qquad (n \geq 1). \tag{5}$$

Step 2: a third-order equation for H. Inserting $G_n = 64^n h_n/\binom{2n}{n}$ into (5) and using $\binom{2n+2}{n+1}/\binom{2n}{n} = 2(2n+1)/(n+1)$ gives the exact three-term recurrence

$$(n+1)^3 h_{n+1} = (2n+1)\Big(n(n+1)+\tfrac{3}{8}\Big)h_n - \frac{n(2n-1)(2n+1)}{4}h_{n-1}. \tag{6}$$

Hence, with $\theta = z\,\mathrm{d}/\mathrm{d}z$, H is annihilated by

$$\theta^3 - \frac{z}{8}(2\theta+1)\big(8\theta(\theta+1)+3\big) + \frac{z^2}{4}(\theta+1)\big(4(\theta+1)^2-1\big) = z\mathcal{S},$$

where

$$\mathcal{S} = s^2\partial_z^3 + 3st\,\partial_z^2 + \Big(t^2 - \tfrac{11}{4}s\Big)\partial_z - \tfrac{3}{8}t. \tag{7}$$

Step 3: symmetric square. B satisfies the Gauss equation $\mathcal{D}B = 0$ with $\mathcal{D} = z(1-z)\partial_z^2 + (1-2z)\partial_z - \frac{3}{16}$, and $\mathcal{S}$ is the corresponding symmetric-square operator: for any f, v one has the exact product identity

$$\mathcal{S}(fv) = sf(\mathcal{D}v)' + (3sf' + tf)\mathcal{D}v + sv(\mathcal{D}f)' + (3sv' + tv)\mathcal{D}f, \tag{8}$$

so $\mathcal{S}(B^2) = 0$. Both H and B^2 have constant term 1, and in (6) the leading coefficient $(n+1)^3$ is invertible in $\mathbb{Q}$; hence a solution of $\mathcal{S}Y = 0$ in $\mathbb{Q}[[z]]$ is uniquely determined by its constant term, and $H = B^2$.

Step 4: the quadratic substitution. The ratio $c_k/c_{k-1} = \frac{(k-\frac{1}{4})(k-\frac{1}{2})(k-\frac{3}{4})}{k^3}$ shows that K is annihilated by

$$\mathcal{M} = w^2(1-w)\partial_w^3 + w\Big(3 - \tfrac{9}{2}w\Big)\partial_w^2 + \Big(1 - \tfrac{51}{16}w\Big)\partial_w - \tfrac{3}{32}.$$

Writing $s = z(1-z)$, $t = 1-2z$ and differentiating $K(w(z))$ formally (using $w' = 4t$, $t^2 = 1-w$) gives the exact operator identity, valid without applying $\mathcal{M}K = 0$,

$$\mathcal{S}_z\big[K(w(z))\big] = 4t\,(\mathcal{M}_w K)(w(z)). \tag{9}$$

Hence $\mathcal{S}(K\circ w) = 0$. Since $K(w(z))$ has constant term 1, the same uniqueness argument gives $H = K\circ w$. The equivalence of the two forms is Clausen's identity ${}_2F_1(\frac{1}{4},\frac{3}{4};1;z)^2 = {}_3F_2(\frac{1}{4},\frac{1}{2},\frac{3}{4};1,1;4z(1-z))$. □

Remark 2.3. The identity $H = B^2$ is of classical Clausen type (compare [3]); we included the operator derivation above because every subsequent argument truncates (9) at degree p, and the truncation error is computed from the same two operators.

For later use we record the finite form of the Clausen step. Let $b_k = \binom{2k}{k}\binom{4k}{2k}/64^k$ and $B_p(t) = \sum_{k<p} b_k t^k$.

Proposition 2.4 (finite Clausen square)**.** *For every odd prime p,*

$$K_p\big(4t(1-t)\big) \equiv B_p(t)^2 \pmod{p^2} \qquad \text{in } \mathbb{Z}_{(p)}[t].$$

Proof. Coefficients of degree $< p$ agree exactly by Theorem 2.2. On the left, terms with $k > (p-1)/2$ vanish modulo p^2 by Lemma 2.1, and $k \le (p-1)/2$ contributes degree $\le p-1$ only. On the right, the coefficient of degree $n \ge p$ is $\sum_{i+j=n} b_i b_j$. For $0 \le i < p$,

$$v_p(b_i) = v_p\Big(\tfrac{(4i)!}{(2i)!}\Big) = \Big\lfloor \tfrac{4i}{p} \Big\rfloor - \Big\lfloor \tfrac{2i}{p} \Big\rfloor = \begin{cases} 0, & i < p/4, \\ 1, & p/4 < i < 3p/4, \\ 2, & 3p/4 < i < p. \end{cases}$$

If $i + j \ge p$ then either one index is $< p/4$ and the other $> 3p/4$, or both indices are $\ge p/4$; in both cases $v_p(b_i b_j) \ge 2$. □

3. The center bridge modulo p^3

Theorem 3.1 (center bridge)**.** *For every odd prime p,*

$$\sum_{n=0}^{p-1} \frac{\binom{2n}{n} G_n}{128^n} \equiv \sum_{k=0}^{p-1} \frac{a_k}{256^k} \pmod{p^3}.$$

Proof. Truncate the identity of Theorem 2.2 at degree $< p$ and then set $z = \frac{1}{2}$. Since $w(z) = 4z(1-z)$ has vanishing order 1, the terms $k \ge p$ of K do not meet the coefficients of degree $< p$, so

$$H_p\Big(\tfrac{1}{2}\Big) = \sum_{k=0}^{p-1} c_k R_{p,k}, \qquad R_{p,k} = \sum_{j=0}^{\min(k,p-1-k)} (-1)^j \binom{k}{j} 2^{k-j} \in \mathbb{Z}. \tag{10}$$

Everything takes place in a finite polynomial ring; no infinite series is evaluated at the degenerate point $w = 1$.

If $2k < p$ the full binomial expansion of $(2-1)^k$ is present and $R_{p,k} = 1$ exactly.

If $2k > p$, write $r = p - 1 - k$, so $0 \le r < k < p$. For $0 \le j \le r$ the factor $j!$ is a p-unit and $k \equiv -1 - r \pmod{p}$, hence

$$\binom{k}{j} \equiv (-1)^j \binom{r+j}{j} \pmod{p}.$$

Combining with the weight 2^{k-j} and the finite identity

$$\sum_{j=0}^{r} \binom{r+j}{j} 2^{-j} = 2^r \qquad \text{in } \mathbb{Q} \tag{11}$$

(proof: if S_r denotes the left side, Pascal's rule gives $S_r = S_{r-1} + 2^{-r}\binom{2r-1}{r} + \frac{1}{2}\big(S_r - 2^{-r}\binom{2r}{r}\big)$, and $\binom{2r}{r} = 2\binom{2r-1}{r}$ forces $S_r = 2S_{r-1}$), we obtain

$$R_{p,k} \equiv 2^{k+r} = 2^{p-1} \equiv 1 \pmod{p}.$$

By Lemma 2.1, the high range $2k > p$ has $v_p(c_k) \ge 2$, so each summand $c_k(R_{p,k} - 1)$ lies in $p^3\mathbb{Z}_{(p)}$; the low-range summands vanish exactly. Summing over k gives the claim, for all odd p including $p = 3$. □

Remark 3.2. The extra factor p at the center has a visible combinatorial origin: the quadratic coordinate $4z(1-z)$, the reflection $k \leftrightarrow p-1-k$ of the truncated binomial expansion, and the carry valuation $\lfloor 4k/p \rfloor$ cooperate. No such extra factor exists at a generic point (Section 4).

The theorem is sharp: the difference is exactly divisible by p^3 in the smallest cases, and its quotient modulo p is as follows.

Theorem 3.3 (first defect at the center)**.** *Let p be an odd prime, $m = (p-1)/2$, and $\Delta_p = p^{-3}\big(H_p(\frac{1}{2}) - K_p(1)\big) \in \mathbb{Z}_{(p)}$. Then*

$$\Delta_p \equiv \sum_{k=(p+1)/2}^{\lfloor (3p-1)/4 \rfloor} \frac{a_k}{p^2}\, 256^{-k} \sum_{j=p-k}^{m} \frac{\binom{2j}{j}}{2j\, 4^j} \pmod{p}. \tag{12}$$

For $p = 3, 5, 7, 11, 13, 17$ the values are $1, 1, 2, 1, 5, 12$ respectively; in particular the center comparison does not lift to p^4 in general.

Proof. For $m < k < p$ write $r = p - 1 - k$. Since $r + i \le 2r < p$, all denominators in

$$(-1)^j \binom{k}{j} = \binom{r+j}{j} \prod_{i=1}^{j} \Big(1 - \frac{p}{r+i}\Big)$$

are p-units, and expanding to first order,

$$R_{p,k} \equiv 2^k\big(2^r - pU_r\big) \pmod{p^2}, \qquad U_r = \sum_{j=0}^{r} \binom{r+j}{j} 2^{-j} (H_{r+j} - H_r),$$

with H_n the harmonic numbers. The polynomial $P_r(X) = \sum_{j=0}^{r} \binom{r+X+j}{j} 2^{-j}$ satisfies $P_r(X) = 2P_{r-1}(X) + 2^{-r}\frac{X}{r+X}\binom{2r+X-1}{r}$ by Pascal's rule; differentiating at $X = 0$ (a finite polynomial operation) gives $U_r = 2U_{r-1} + \binom{2r}{r}/(r2^{r+1})$, hence the exact identity

$$2^{-r} U_r = L_r := \sum_{j=1}^{r} \frac{\binom{2j}{j}}{2j\, 4^j}.$$

Therefore, with $q_p(2) = (2^{p-1} - 1)/p$,

$$\frac{R_{p,k} - 1}{p} \equiv q_p(2) - L_r \pmod{p}. \tag{13}$$

At the boundary $k = r = m$ one has $R_{p,m} = 1$ exactly, whence $q_p(2) \equiv L_m \pmod{p}$ and the bracket in (13) is $L_m - L_r = \sum_{j=r+1}^{m} \binom{2j}{j}/(2j4^j)$. Finally $\Delta_p = \sum_{k=m+1}^{p-1} \frac{a_k}{p^2} 256^{-k} \frac{R_{p,k}-1}{p}$; by Lemma 2.1, $\frac{a_k}{p^2}$ is integral, and it is divisible by p once $4k \ge 3p$ (equality $4k = 3p$ is impossible), so exactly the range $m < k$, $4k < 3p$ survives, giving (12). The listed values are obtained by direct exact evaluation. □

4. The Euler transfer modulo p^2 and the pairing

Write

$$D_p(z) = H_p(z) - K_p(w(z)) \in \mathbb{Q}[z], \qquad w(z) = 4z(1-z).$$

Theorem 4.1 (weighted transfer and first defect)**.** *For every odd prime p,*

$$D_p(z) = -\sum_{k=(p+1)/2}^{p-1} a_k 64^{-k} \sum_{j=p-k}^{k} (-1)^j \binom{k}{j} z^{k+j},$$

hence $D_p \in p^2\mathbb{Z}_{(p)}[z]$. Moreover, with $b = \lfloor (3p-1)/4 \rfloor$,

$$D_p(z) \equiv p^2 z^p R_p(z) \pmod{p^3},$$

$$R_p(z) = -\sum_{k=(p+1)/2}^{b} \frac{a_k}{p^2}\, 64^{-k} \sum_{j=p-k}^{k} (-1)^j \binom{k}{j} z^{k+j-p}, \tag{14}$$

and, viewing the reductions $\bar{R}_p, \bar{R}'_p \in \mathbb{F}_p[z]$,

$$\bar{R}_p(1-z) = -\bar{R}_p(z), \qquad \bar{R}'_p(1-z) = \bar{R}'_p(z) \qquad \text{in } \mathbb{F}_p[z]. \tag{15}$$

Proof. By Theorem 2.2, the coefficients of degree $< p$ of $H - K \circ w$ vanish, so D_p receives contributions only from terms of $K \circ w$ with $k > (p-1)/2$; expanding $(4z)^k(1-z)^k$ and discarding degrees $< p$ gives the displayed formula (note $j + k \geq p$ is automatic there). The integrality $D_p \in p^2\mathbb{Z}_{(p)}[z]$ is immediate from Lemma 2.1, including $p = 3$. Reducing (14): terms with $v_p(a_k) \geq 3$ (i.e. $k > b$) are 0 modulo p^3, and the top surviving exponent is $2b - p < p$, so R_p has degree $2\lfloor (p-3)/4 \rfloor + 1 < p$.

For the reflection (15) we use the following elementary projection lemma, applied repeatedly: a polynomial $T \in \mathbb{F}_p[z]$ of degree $< 2p$ decomposes *uniquely* as

$$T(z) = L(z) + z^p V(z), \qquad \deg L < p,\ \deg V < p, \tag{16}$$

simply by splitting its monomials according to whether the exponent is $< p$ or $\geq p$; if moreover $T(1-z) = T(z)$, then comparing the coefficients of degree $\geq p$ in $T(1-z) = T(z)$ and using $(1-z)^p = 1 - z^p$ in $\mathbb{F}_p[z]$ gives $V(1-z) = -V(z)$, so that $T(1-z) - T(z) = -2z^p V(z)$ in $\mathbb{F}_p[z]$.

Apply this to the reduction $\bar{D}_p \in \mathbb{F}_p[z]$ of D_p/p^2, which has degree $\leq 2b - p + 2(p-1) < 2p$ by the degree bound above. Since $w(1-z) = w(z)$, the polynomial $K_p \circ w$ is invariant under $z \mapsto 1-z$; in $\mathbb{F}_p[z]$ each term $z^k(1-z)^k$ of $\bar{D}_p$ (coming from (14) modulo p, i.e. from $z^k(1-z)^k$ expanded at degrees $\geq p$) is itself invariant, hence so is $\bar{D}_p$; and H_p is invariant only in the combination $H_p(z) - H_p(1-z)$, so writing $\bar{D}_p = L + z^p V$ as in (16) the invariance of each tail chunk $[z^k(1-z)^k]_{\geq p} = z^p V_k(z)$ gives $V_k(1-z) = -V_k(z)$ for every k, hence $V(1-z) = -V(z)$. But by (14) the reduction of V modulo p is exactly $\bar{R}_p$, the reduction of R_p. Therefore $\bar{R}_p(1-z) = -\bar{R}_p(z)$ in $\mathbb{F}_p[z]$, which is the first identity of (15). Differentiating it gives the second. (Note that (15) is genuinely a statement over $\mathbb{F}_p$: the first defect R_p enters

D_p only through the p^2-multiple in (14), so only its reduction modulo p is defined intrinsically.) □

Corollary 4.2 (reflection-difference form)**.** *In* $\mathbb{Z}_{(p)}[z]$*,*

$$H_p(z) - H_p(1-z) \equiv p^2 R_p(z) \pmod{p^3}. \tag{17}$$

Proof. Since $w(1-z) = w(z)$ we have $H_p(z) - H_p(1-z) = D_p(z) - D_p(1-z)$ exactly. By (14),

$$D_p(z) - D_p(1-z) \equiv p^2\big(z^p R_p(z) - (1-z)^p R_p(1-z)\big) \pmod{p^3}.$$

Reducing the right-hand factor modulo p and using $(1-z)^p \equiv 1 - z^p$ together with $R_p(1-z) \equiv -R_p(z) \pmod{p}$ from (15) gives $z^p R_p(z) - (1-z)^p R_p(1-z) \equiv R_p(z) \pmod{p}$, and since the congruence is p^2 times a quantity computed modulo p, this yields (17). □

Corollary 4.3 (pairing)**.** *Let* $\mathcal{W}_p(z) = zH_p'(z) = \sum_{n<p} n h_n z^n$*. Then in* $\mathbb{Z}_{(p)}[z]$*,*

$$(1-z)\mathcal{W}_p(z) + z\mathcal{W}_p(1-z) \equiv p^2 z(1-z)R_p'(z) \pmod{p^3}, \tag{18}$$

and whenever $z, 1-z$ *are p-units,*

$$\mathcal{W}_p(z) \equiv \frac{1-2z}{1-z}\,(\theta_w K_p)\big(w(z)\big) \pmod{p^2}, \qquad \theta_w = w\frac{\mathrm{d}}{\mathrm{d}w}. \tag{19}$$

Proof. Differentiating the congruence (17) of Corollary 4.2 (differentiation is a well-defined operation on coefficientwise congruences of polynomials) gives $\mathcal{W}_p(z)/z + \mathcal{W}_p(1-z)/(1-z) \equiv p^2 R_p'(z) \pmod{p^3}$; multiplying by $z(1-z)$ yields (18). For (19), differentiating $H_p \equiv K_p \circ w \pmod{p^2}$ and using $\theta_z w = 4z(1-2z)$, $w/z = 4(1-z)$ gives $\mathcal{W}_p(z) \equiv \frac{1-2z}{1-z}(\theta_w K_p)(w(z)) \pmod{p^2}$. □

Remark 4.4 (sharpness). The mod p^2 pairings of the table below do not lift to $\equiv 0$ modulo p^3 in general: at $p = 5$ one computes $R_5 = z + 2$, so $R_5' \equiv 1$, and the three denominator-cleared pairings $\mathcal{W}_5(72) + 8\mathcal{W}_5(576)$, $\mathcal{W}_5(48) - 4\mathcal{W}_5(-192)$, $\mathcal{W}_5(63) - 64\mathcal{W}_5(-4032)$ are congruent to $50, 75, 75 \pmod{125}$ respectively (direct evaluation), matching Remark 4.5; all are nonzero. (Exact values in Appendix A.)

The three point pairs of Sun's conjecture arise as follows. Writing

$$\mathcal{W}_p(M) := \sum_{n<p} n\binom{2n}{n} G_n / M^n = \mathcal{W}_p(64/M),$$

and recalling $U_p(d) = \sum_{k<p} k a_k / d^k$ from (3), one has $w(z)/256 = 1/d$ and the Euler factors $\frac{1-2z}{1-z} \in \{-7,\, 5,\, 65\}$:

$(M, 64/(1-z))$	$(z, 1-z)$	$d = 256/w$	Euler	pairing $(\mathrm{mod}\ p^2)$
$(72, 576)$	$(\frac{8}{9}, \frac{1}{9})$	648	-7	$\mathcal{W}_p(72) + 8\mathcal{W}_p(576) \equiv 0$
$(48, -192)$	$(\frac{4}{3}, -\frac{1}{3})$	-144	5	$\mathcal{W}_p(48) - 4\mathcal{W}_p(-192) \equiv 0$
$(63, -4032)$	$(\frac{64}{63}, -\frac{1}{63})$	-3969	65	$\mathcal{W}_p(63) - 64\mathcal{W}_p(-4032) \equiv 0$

The first two pairings hold for $p>3$; the third requires also $p\neq 7$ (both because 63 must be a p-unit and because $1-z=-1/63$ appears). In each case $1-z$ is a p-unit in the stated range, so (19) gives

$$\begin{aligned}\mathcal{W}_p(72)&\equiv -7U_p(648),\qquad \mathcal{W}_p(48)\equiv 5U_p(-144),\\ \mathcal{W}_p(63)&\equiv 65U_p(-3969)\pmod{p^2}.\end{aligned}\tag{20}$$

Remark 4.5 (mod p^3 refinement of the pairings). Clearing denominators in (18) at the three points gives the exact first defects of the paired sums:

$$\begin{aligned}\mathcal{W}_p(72)+8\mathcal{W}_p(576)&\equiv \frac{8p^2}{9}\,R_p'\Big(\frac{8}{9}\Big),\\ \mathcal{W}_p(48)-4\mathcal{W}_p(-192)&\equiv \frac{4p^2}{3}\,R_p'\Big(\frac{4}{3}\Big),\\ \mathcal{W}_p(63)-64\mathcal{W}_p(-4032)&\equiv \frac{64p^2}{63}\,R_p'\Big(\frac{64}{63}\Big)\end{aligned}$$

modulo p^3 (in the ranges of the table, all displayed denominators being p-units). At $p=5$, where $R_5=z+2$, the three right-hand sides evaluate to $50,75,75 \pmod{125}$ (cf. Remark 4.4); in particular the mod p^2 pairings are sharp.

5. Structure of the defect polynomial

This section is not needed for the proof of the conjecture, but it explains *why* the first defect is manageable: R_p satisfies a constant-coefficient forcing equation and is carried by a kernel of about half its apparent degree. Let $\varepsilon=\left(\frac{-1}{p}\right)$ and let $\mathcal{S}$ be the operator (7).

Theorem 5.1. *Let p be an odd prime and $q=\lfloor (p-3)/4\rfloor$.*

(i) *Forcing equation: $\mathcal{S}R_p=\varepsilon/4$ in $\mathbb{F}_p[z]$.*

(ii) *Kernel: $R_p(z)=t\,J_p(w)$ with $t=1-2z$, $w=4z(1-z)$, where*

$$J_p(w)=-\frac{\varepsilon}{2}\sum_{j=0}^{q}\frac{(\frac12)_j(\frac34)_j(\frac54)_j}{(\frac32)_j^3}\sum_{h=0}^{j}\frac{(-j)_h}{(\frac12-j)_h}w^h,\tag{21}$$

all denominators being p-units.

(iii) *Integral invariant: with $a=\lfloor p/4\rfloor$, $F=P_a(1-2z)$ the Legendre polynomial part and $\lambda=a(a+1)$, the expression*

$$I_F[Y]=2s^2(F')^2Y-2s^2FF'Y'+s^2F^2Y''+stF^2Y'+2\lambda sF^2Y$$

satisfies $I_F[Y]'=F^2\mathcal{S}Y$ identically, and

$$I_F[R_p]=\frac{\varepsilon}{4}\Big(\int_0^z F(u)^2\mathrm{d}u-\frac{z^p}{2a+1}\Big)\qquad \text{in } \mathbb{F}_p[z].$$

The term $z^p/(2a+1)$ is forced by the fact that in characteristic p the kernel of $\mathrm{d}/\mathrm{d}z$ on polynomials of degree $\le p$ is spanned by 1 and z^p; it

is pinned down by $I_F[R_p](0) = I_F[R_p](1) = 0$ *and the Legendre square integral* $\int_0^1 F^2 = 1/(2a+1)$.

Proof. (i) Applying $\mathcal{S}$ to the truncations gives the exact boundary computations

$$\mathcal{S}H_p = -p^3 h_p z^{p-1} + \frac{p(4p^2-1)}{4} h_{p-1} z^p, \qquad \mathcal{M}K_p = -p^3 c_p w^{p-1},$$

and the pullback (9) holds for the truncations up to the same boundary terms. Since $v_p(h_p), v_p(c_p) \geq 1$, reducing $\mathcal{S}(H_p - K_p \circ w) \equiv p^2 \mathcal{S}(z^p R_p) \pmod{p^3}$ gives $\mathcal{S}(z^p R_p) = -(h_{p-1}/p) z^p/4$ in $\mathbb{F}_p[z]$. In characteristic p one has $\mathcal{S}(z^p Y) = z^p \mathcal{S} Y$ (every term of $\mathcal{S}$ except the zeroth-order one contains a derivative), so cancelling z^p leaves $\mathcal{S}R_p = -(h_{p-1}/p)/4$. Finally $h_{p-1}/p \equiv -\varepsilon \pmod{p}$: in the defining convolution for G_{p-1}, every off-central summand contains a factor p in at least one of its central binomial coefficients. The sole surviving term has $k = m = (p-1)/2$; using $\binom{p-1}{m} \equiv (-1)^m$ and $4^m \equiv (-1)^m$ shows that it equals $(-1)^m = \varepsilon$ modulo p. Thus $G_{p-1} \equiv \varepsilon \pmod{p}$, and $\binom{2p-2}{p-1}/p \equiv -1$ together with $64^{p-1} \equiv 1$ yields the sign.

(ii) The reflection in (15) and $t^2 = 1 - w$ first give a unique factorization $R_p = t J_p(w)$ with $\deg J_p \leq q$. To identify J_p, put $m = (p-1)/2$, write $k = m + 1 + j$ and $r = p - 1 - k = m - 1 - j$, and set

$$V_{p,k}(z) = z^{-p}[z^k (1-z)^k]_{\geq p}.$$

The coefficient of z^{p-1} in the complementary low part is $(-1)^r \binom{k}{r} \equiv \binom{2r}{r} =: B_r$. Since the complete polynomial $P = z^k(1-z)^k$ satisfies $sP' - ktP = 0$, separating its low and high parts gives the exact equation

$$sV'_{p,k} - ktV_{p,k} = -(2k+1)B_r. \tag{5.3}$$

Writing $V_{p,k} = tQ_j(w)$ and using $k \equiv j + \frac{1}{2} \pmod{p}$ turns this into

$$w(1-w)Q'_j + (jw - j - \tfrac{1}{2})Q_j = -(2j+2)B_r.$$

Its coefficient recursion, beginning with $Q_j(0) = (2k+1)B_r/k$, has the polynomial solution

$$Q_j(w) = \frac{2k+1}{k} B_r \sum_{h=0}^{j} \frac{(-j)_h}{(\frac{1}{2} - j)_h} w^h. \tag{5.4}$$

All denominators here are p-units. The scalar of this tail in (14) is $u_j = (a_{m+1+j}/p^2) 64^{-(m+1+j)}$. Wilson's theorem gives $u_0 = 1$ and

$$\frac{u_j}{u_{j-1}} = \frac{2j(16j^2-1)}{(2j+1)^3}, \qquad \frac{((2k+1)/k)B_r}{((2k-1)/(k-1))B_{r+1}} = \frac{2j-1}{8j}.$$

Their product is $\frac{(j-1/2)(j-1/4)(j+1/4)}{(j+1/2)^3}$; the initial value is $\varepsilon/2$. Substitution in the defining tail for R_p proves (21). Its top coefficient is nonzero, so $\deg J_p = q$ and $\deg R_p = 2q+1$.

(iii) The identity $I_F[Y]' = F^2\mathcal{S}Y$ is a formal consequence of $\mathcal{D}F = 0$ via the product rules. Integrating (i) termwise (legitimate since $2a+1 < p$, so no denominator p occurs) gives

$$I_F[R_p] = \frac{\varepsilon}{4}\int_0^z F(u)^2 \mathrm{d}u + C_0 + C_1 z^p \qquad \text{in } \mathbb{F}_p[z],$$

where the integration ambiguity is $C_0 + C_1 z^p$ because in characteristic p the kernel of $\mathrm{d}/\mathrm{d}z$ on polynomials of degree $\le p$ is spanned by 1 and z^p. To pin down the constants we use the reflections: $F(1-z) = (-1)^a F(z)$, and each of the five terms of $I_F[Y]$ is built from an even number of F's or F''s together with the reflection-odd R_p (and $s = z(1-z)$ invariant, $t = 1-2z$ odd), so $I_F[R_p](1-z) = -I_F[R_p](z)$. Evaluation at $z = 0$ gives $I_F[R_p](0) = 0$ (every term contains s or $R_p(0) = 0$), hence $C_0 = 0$; evaluation at $z = 1$ then gives $0 = I_F[R_p](1) = \frac{\varepsilon}{4}\int_0^1 F^2 + C_1$, and the classical Legendre square integral (Rodrigues formula and integration by parts, valid in $\mathbb{F}_p$ since $2a+1 < p$) $\int_0^1 F(u)^2 \mathrm{d}u = 1/(2a+1)$ yields $C_1 = -\varepsilon/(4(2a+1))$. □

Remark 5.2. The forcing equation (i) with the reflection (15) characterizes R_p uniquely among polynomials of degree $< p$: the homogeneous solution of $\mathcal{S}Y = 0$ regular at 0 is F^2, which is reflection-*even*, while the recurrence for the coefficients is determined by the constant term. Formula (21) and the invariant (iii) are two cross-checkable presentations of the same defect.

6. Special values: the three weighted pairs

We now combine (20) with published special values of the triple binomial sums (3). The small primes excluded by the printed hypotheses are settled by direct exact computation (Appendix A).

6.1. The pair $(72, 576)$.

Theorem 6.1. *For every prime $p > 3$,*

$$\mathcal{W}_p(72) \equiv -8\mathcal{W}_p(576) \equiv \begin{cases} 4x^2 - 3p, & p = x^2 + 4y^2 \equiv 1 \pmod 4, \\ p, & p \equiv 3 \pmod 4, \end{cases} \pmod{p^2}.$$

Proof. The pairing is Corollary 4.3. For $p \neq 7$, Sun [13, Theorem 2.10] gives (the printed hypotheses exclude $p = 2, 3, 7$)

$$U_p(648) \equiv \begin{cases} (3p - 4x^2)/7 \pmod{p^2}, & p = x^2 + 4y^2 \equiv 1 \pmod 4, \\ -p/7 \pmod{p^2}, & p \equiv 3 \pmod 4, \end{cases}$$

and (20) with factor -7 yields the values. For $p = 7$: direct evaluation from (1) gives $\mathcal{W}_7(72) \equiv 7 \pmod{49}$ and $\mathcal{W}_7(72) + 8\mathcal{W}_7(576) \equiv 0 \pmod{49}$ (Appendix A), which is the $p \equiv 3 \pmod 4$ branch. □

6.2. **The pair** $(48,-192)$**.**

Theorem 6.2. *For every prime $p>3$,*

$$\mathcal{W}_p(48)\equiv 4\mathcal{W}_p(-192)\equiv\begin{cases}3p-4x^2, & p=x^2+3y^2\equiv 1 \quad (3),\\ -p, & p\equiv 2 \quad (3),\end{cases}\pmod{p^2}.$$

Proof. For $p>5$, Sun [13, Theorem 2.9] gives $U_p(-144)$ modulo p^2 with values $(3p-4x^2)/5$ and $-p/5$ in the split and inert cases; multiplication by 5 via (20) gives the claim. The printed range of that theorem is $p>3$, but the formula containing $1/5$ does not hold at $p=5$ as stated: $U_5(-144)\equiv 19 \pmod{25}$ while $-p/5\equiv -1\equiv 24 \pmod{25}$. We therefore quote it for $p>5$ only and settle $p=5$ directly: $\mathcal{W}_5(48)\equiv 20\equiv -5 \pmod{25}$ and $\mathcal{W}_5(48)-4\mathcal{W}_5(-192)\equiv 0 \pmod{25}$ (Appendix A), the inert branch. □

6.3. **The pair** $(63,-4032)$**: a CM Ramanujan sum.** By (20), the third pair reduces to $65U_p(-3969)$. The required input is the following explicit specialization of the general CM theorem of Chisholm–Deines–Long–Nebe–Swisher [4, Theorem 1] (restated with both reduction types in [2, Theorem 2.1]).

Proposition 6.3. *For every prime $p>3$, $p\neq 7$,*

$$\sum_{k=0}^{p-1}(65k+8)\frac{a_k}{(-3969)^k}\equiv 8\left(\frac{-7}{p}\right)p \pmod{p^2}. \tag{22}$$

Proof. Parameters. Since $\frac{(1/2)_k(1/4)_k(3/4)_k}{(k!)^3}=\frac{a_k}{256^k}$, the sum (22) is 8 times the normalized truncated sum with

$$d=4,\qquad \lambda=-\frac{256}{3969},\qquad \alpha=\frac{65}{8},\qquad \delta=\frac{9\sqrt{7}}{8};$$

the identity $\sum_{k\geq 0}(65k+8)a_k/(-3969)^k=9\sqrt{7}/\pi$ [5, Table 2] certifies that these are the parameters of a Ramanujan-type formula in the sense required by the cited theorem. (An independent normalization check: Clausen–Pfaff gives $K(-4x/(1-x)^2)=\sqrt{1-x}\,F(x)$ between the ${}_3F_2$ with parameters $(\frac14,\frac12,\frac34)$ and $(\frac12,\frac12,\frac12)$, and at $x=1/64$ differentiation gives $(65\theta_w+8)K=\frac{9\sqrt{7}}{16}(42\theta_x+5)F$; the classical value $(42k+5)$-series $=16/\pi$ [9, (1.1)] gives the same δ.)

Hypotheses. $1-\lambda=(65/63)^2$, so $\mathbb{Q}(\sqrt{1-\lambda})=\mathbb{Q}$ (totally real, unramified conditions automatic) and $|\lambda|<1$. With $t=(1-\sqrt{1-\lambda})/2=-1/63$, the curve of the theorem is

$$E:\ y^2=x\Big(x^2+x-\tfrac{1}{252}\Big),\qquad \Delta(E)=\frac{2^6}{3^6 7^3},\qquad j(E)=255^3=16581375.$$

The j-invariant is the singular modulus of the CM order of discriminant -28 (the class polynomial is $H_{-28}(X)=X-16581375$; compare [16]), so E has CM by $\mathbb{Q}(\sqrt{-7})$; for $p>3$, $p\neq 7$ the displayed model has p-integral

coefficients and unit discriminant, hence good reduction. By Deuring's criterion the ordinary (resp. supersingular) primes are the split (resp. inert) primes of $\mathbb{Q}(\sqrt{-7})$, so the sign in the theorem is $\mathrm{sgn}(E,p) = \left(\frac{-7}{p}\right)$. For $p \notin \{2,3,5,7,13\}$ both α and λ are p-units and $\left(\frac{1-\lambda}{p}\right) = 1$, so [4, Theorem 1] applies verbatim and multiplication by 8 gives (22) — including the inert classes $p \equiv 3,5,6 \pmod 7$, for which the theorem assigns $-p$ in the normalized sum (we do not use any ordinary-prime-only variant).

The two exceptional primes. At $p = 5, 13$ the parameter $\alpha = 65/8$ is not a p-unit and $p^2 \mid 1-\lambda$, so the theorem cannot be invoked. Direct exact evaluation of the cleared-denominator sum (no division by 65 is ever performed) gives

p	$\big((65k+8)a_k(-3969)^{-k} \bmod p^2\big)_{0\le k<p}$	sum
5	8, 17, 10, 0, 0	$10 \equiv -40 \pmod{25}$
13	8, 160, 142, 80, 52, 65, 65, 0, …, 0	$65 \equiv -104 \pmod{169}$

Both primes are inert in $\mathbb{Q}(\sqrt{-7})$, and the sums equal $-8p$ modulo p^2, as required. (Terms with $k > (p-1)/2$ vanish modulo p^2 by Lemma 2.1.) □

Remark 6.4 (a period-product form). Proposition 2.4 allows formal differentiation modulo p^2: at $t_0 = -1/63$ the chain rule reads $65\theta_w = 64\theta_t$, and with B_p as in Section 2,

$$\sum_{k<p}(65k+8)\frac{a_k}{(-3969)^k} \equiv 8B_p(t_0)\big(B_p(t_0) + 16\,\theta_t B_p(t_0)\big) \pmod{p^2}$$

for $p \nmid 2\cdot3\cdot5\cdot7\cdot13$; the two factors are the finite period and its Gauss–Manin derivative of E. This gives a concrete geometric reading of (22); it is not used in the proof.

Theorem 6.5. *For every prime $p > 3$, $p \neq 7$,*

$$\mathcal{W}_p(63) \equiv 64\mathcal{W}_p(-4032) \pmod{p^2},$$

with common value

$$\begin{cases} -8(4x^2-3p), & p = x^2+7y^2 \equiv 1,2,4 \quad (7), \\ -8p, & p \equiv 3,5,6 \quad (7). \end{cases}$$

Proof. Write $T_p = T_p(-3969)$, $U_p = U_p(-3969)$. Proposition 6.3 reads $65U_p + 8T_p \equiv 8\left(\frac{-7}{p}\right)p$. For split p ($p = x^2+7y^2 \equiv 1,2,4 \pmod 7$), Sun–Ye [15, Theorem 1.8] give $T_p \equiv 4x^2 - 2p \pmod{p^2}$ (in fact modulo p^3); for inert p ($p \equiv 3,5,6 \pmod 7$, $p \neq 7$), Sun [11, Theorem 5.1] gives $T_p \equiv 0 \pmod{p^2}$. Hence

$$65U_p \equiv 8\left(\frac{-7}{p}\right)p - 8T_p \equiv \begin{cases} 8p - 8(4x^2-2p) = -8(4x^2-3p), & \text{split}, \\ -8p, & \text{inert}, \end{cases}$$

and (20) identifies the left side with $\mathcal{W}_p(63) \equiv 64\mathcal{W}_p(-4032) \pmod{p^2}$. □

7. The denominator 128 modulo p^3

Theorem 7.1 (split branch)**.** *For $p = x^2 + 2y^2 \equiv 1, 3 \pmod 8$,*

$$\sum_{n=0}^{p-1} \frac{\binom{2n}{n} G_n}{128^n} \equiv 4x^2 - 2p - \frac{p^2}{4x^2} \pmod{p^3}.$$

Proof. Theorem 3.1 reduces the left side to $T_p = \sum_{k<p} a_k/256^k$ modulo p^3, and Sun–Ye [15, Theorem 1.5] evaluate exactly this sum: $T_p \equiv 4x^2 - 2p - p^2/(4x^2) \pmod{p^3}$ for all $p = x^2+2y^2 \equiv 1, 3 \pmod 8$ (no excluded primes). Note $p = 3 = 1^2 + 2 \cdot 1^2$ is covered; the central binomial $\binom{2n}{n} G_n/128^n$ is p-integral throughout. □

For the inert classes we use the jet comparison proved in Appendix B and the following exact rational identity, proved in Appendix C. Let, for $a \geq 1$,

$$f_i = (-1)^i \binom{a}{i} \binom{a+i}{i},$$

$$\mathcal{Q}_a(z) = \sum_{j=0}^{a-1} \sum_{k=0}^{a-1-j} f_{k+j+1}\, (-1)^k \frac{(a-k-1)!\,(k!)^2}{(a+k+1)!}\, z^j.$$

Theorem 7.2. *For every odd $a \geq 1$,*

$$\mathcal{Q}_a\Big(\tfrac{1}{2}\Big) = -\Big(\frac{2^{a-1}}{a\binom{a-1}{(a-1)/2}}\Big)^2.$$

Proposition 7.3 (jet reduction at the center)**.** *Let $p \geq 5$ and $a = \lfloor p/4 \rfloor$. Then $H_p(\frac{1}{2}) \equiv A_p(\frac{1}{2})^2 + \frac{3p^2}{16}\mathcal{Q}_a(\frac{1}{2}) \pmod{p^3}$, where A_p is the truncation of B. If a is odd (equivalently $p \equiv 5, 7 \pmod 8$), then $A_p(\frac{1}{2}) \in p^2\mathbb{Z}_{(p)}$, so*

$$\sum_{n=0}^{p-1} \frac{\binom{2n}{n} G_n}{128^n} \equiv \frac{3p^2}{16}\mathcal{Q}_a\Big(\tfrac{1}{2}\Big) \pmod{p^3}.$$

Theorem 7.4 (inert branches)**.** *Let $p > 3$ be prime and $C_p = \binom{\lfloor p/4 \rfloor}{\lfloor p/8 \rfloor}$. Then*

$$\sum_{n=0}^{p-1} \frac{\binom{2n}{n} G_n}{128^n} \equiv \begin{cases} \dfrac{p^2}{3} C_p^{-2} \pmod{p^3}, & p \equiv 5 \pmod 8, \\ -\dfrac{3p^2}{2} C_p^{-2} \pmod{p^3}, & p \equiv 7 \pmod 8. \end{cases}$$

Proof. Write $p = 8k + 5$ or $8k + 7$; then $a = \lfloor p/4 \rfloor = 2k + 1$, $\lfloor p/8 \rfloor = k$, and $C_p = \binom{a}{k}$, a p-unit. Theorem 7.2 gives $\mathcal{Q}_a(\frac{1}{2}) = -16^k/((k+1)^2\binom{a}{k}^2)$. If $p = 8k + 5$: $16^k = 2^{(p-5)/2} = 2^{(p-1)/2}/4 \equiv \left(\frac{2}{p}\right)/4 = -\frac{1}{4}$ and $k + 1 \equiv \frac{3}{8} \pmod p$; if $p = 8k + 7$: $16^k \equiv \frac{1}{8}$ and $k + 1 \equiv \frac{1}{8}$. Substituting into Proposition 7.3:

$$\frac{3}{16}\mathcal{Q}_a\Big(\tfrac{1}{2}\Big) \equiv \begin{cases} \frac{3}{16} \cdot \frac{1/4}{(3/8)^2} C_p^{-2} = \frac{1}{3} C_p^{-2}, & p \equiv 5 \pmod 8, \\ \frac{3}{16} \cdot \frac{-1/8}{(1/8)^2} C_p^{-2} = -\frac{3}{2} C_p^{-2}, & p \equiv 7 \pmod 8, \end{cases}$$

and multiplication by p^2 gives the claim. □

8. Conclusion and remarks

Theorem 8.1. *Conjecture 2.4 of* [12] *holds: the three weighted pairs are Theorems 6.1, 6.2, 6.5, and the denominator*-128 *congruences are Theorems 7.1 and 7.4.*

Remark 8.2 (higher precision at the center). Theorem 3.3 shows that the center bridge lifts to p^4 exactly at the primes where the finite sum (12) vanishes modulo p; the values $1, 1, 2, 1, 5, 12$ at $p = 3, 5, 7, 11, 13, 17$ show this is not automatic. We have no criterion for the vanishing in general.

Remark 8.3 (no naive finite Pfaff transplant). One might hope to convert (22) into the classical $42k + 5$ truncated sum via a direct finite transformation. This is impossible without a correction term: the known modulo-p^2 value $\sum_{k<p}(42k + 5)\binom{2k}{k}^3/4096^k \equiv 5\left(\frac{-1}{p}\right)p$ (compare [9, Theorem 1.1]) is incompatible with (22) under the infinite-series prefactor $9\sqrt{7}/16$ unless $p \mid 47^2$. Any finite analogue of the Clausen–Pfaff transformation between these two truncations must therefore contain a first-order tail correction; we have not constructed one.

Remark 8.4 (scope). The same two-level transfer applies in principle to other CM points of the family, provided the corresponding special values of the triple binomial sums are known; the defect analysis of Section 5 is point-free. We have not pursued further instances.

Appendix A. Exceptional primes: exact certificates

All computations below are exact integer/rational evaluations of the defining sums (1), (3); they are finite certificates covering exactly the primes excluded by the quoted theorems, not samples suggesting a general pattern.

Group $(72, 576)$ **at** $p = 7$**.** $\mathcal{W}_7(72) \equiv 7 \pmod{49}$ and $\mathcal{W}_7(576) \equiv 0 \pmod{49}$; in particular the pairing and the value p hold.
Group $(48, -192)$ **at** $p = 5$**.** $\mathcal{W}_5(48) \equiv 20 \equiv -5 \pmod{25}$ and $\mathcal{W}_5(-192) \equiv 0 \pmod{25}$; the pairing and the value $-p$ hold. For comparison, the printed $p > 3$ formula of [13, Theorem 2.9] would give $U_5(-144) \equiv -1 \pmod{25}$, while the exact value is 19; the corrected combination $5U_5(-144)+T_5(-144)$ is still $\equiv -5 \pmod{25}$.
Proposition 6.3 at $p = 5, 13$**.** The full residue tables are displayed in the proof of Proposition 6.3; the sums are $10 \equiv -40 \pmod{25}$ and $65 \equiv -104 \pmod{169}$, matching $8\left(\frac{-7}{p}\right)p$ since both primes are inert in $\mathbb{Q}(\sqrt{-7})$.
Center at $p = 3$**.** $\sum_{n<3}\binom{2n}{n}G_n/128^n \equiv \sum_{k<3} a_k/256^k \equiv 16 \pmod{27}$; the exact difference is $945/8192 = 3^3 \cdot 35/8192$ with $\Delta_3 = 35/8192 \equiv 1 \pmod 3$, matching (12).

Pairing defects at $p = 5$ (cf. Remark 4.4):

$$\begin{aligned}\mathcal{W}_5(72) + 8\mathcal{W}_5(576) &\equiv 50,\\ \mathcal{W}_5(48) - 4\mathcal{W}_5(-192) &\equiv 75,\\ \mathcal{W}_5(63) - 64\mathcal{W}_5(-4032) &\equiv 75\end{aligned}$$

modulo 125.

Appendix B. The parametric jet and the corrected reflection

This appendix proves Proposition 7.3. Fix $p \geq 5$, $a = \lfloor p/4 \rfloor$, $b = p - a - 1$. Let $B_n = \binom{2n}{n}\binom{4n}{2n}$, $A_p(z) = \sum_{n<p} B_n z^n/64^n$, and define the finite Pochhammer polynomial

$$J_p(\nu, z) = \sum_{n=0}^{p-1} \frac{(-\nu)_n(\nu+1)_n}{(n!)^2} z^n \in \mathbb{Q}(\nu)[z],$$

whose Taylor coefficients at $\nu = a$ are p-integral since each $n!$ is a p-unit. Write $J_p(a+s, z) = F(z) + sU(z) + s^2V(z) + O(s^3)$.

Lemma B.1. *$F(z) = P_a(1-2z)$ has degree a, and $A_p \equiv F - \frac{p}{4}U + \frac{p^2}{16}V \pmod{p^3}$ in $\mathbb{Z}_{(p)}[z]$.*

Proof. At $\nu = a$,

$$\frac{(-\nu)_n(\nu+1)_n}{(n!)^2} = (-1)^n \frac{(a+n)!}{(a-n)!(n!)^2} qquad (0 \leq n \leq a),$$

which is the coefficient of $P_a(1-2z)$; for $n > a$ the factor $(-\nu)_n$ vanishes. Thus $F = P_a(1-2z)$ exactly. Now

$$a - \frac{p}{4} = \begin{cases} -\frac{1}{4}, & p \equiv 1 \pmod 4,\\ -\frac{3}{4}, & p \equiv 3 \pmod 4,\end{cases}$$

and in the two cases the upper parameters $(-(a - p/4), a - p/4 + 1)$ are respectively $(\frac{1}{4}, \frac{3}{4})$ and $(\frac{3}{4}, \frac{1}{4})$. Consequently

$$A_p(z) = J_p(a - p/4, z)$$

coefficientwise, with no analytic continuation involved. Taylor's formula at the integral point a therefore gives

$$A_p = F - \frac{p}{4}U + \frac{p^2}{16}V + p^3 Z$$

for some $Z \in \mathbb{Z}_{(p)}[z]$: every coefficient of J_p is a polynomial in ν over $\mathbb{Z}_{(p)}$, because $n!$ is a p-unit for $n < p$. This proves the asserted congruence and also fixes the signs and the normalization of V (the coefficient of s^2, not the second derivative itself). □

Lemma B.2. *$[z^n]U$ is given, with $f_n = (-1)^n\binom{a}{n}\binom{a+n}{n}$, by*

$$[z^n]U = \begin{cases} f_n(H_{a+n} - H_{a-n}), & 0 \le n \le a, \\ (-1)^{a+1}(n-a-1)!\,(n+a)!/(n!)^2, & a < n < p, \end{cases}$$

and $D_p := U \bmod p$ has degree b. With $W_p(z) = \sum_{n=b+1}^{p-1}\big([z^n]U/p\big)z^n \bmod p$, Wilson's theorem rewrites it as the short reversed polynomial

$$W_p(z) = \sum_{k=0}^{a-1} w_k z^{p-1-k}, \qquad w_k = (-1)^k \frac{(a-k-1)!(k!)^2}{(a+k+1)!}.$$

Proof. For $n \le a$, logarithmic differentiation of both Pochhammer factors at $\nu = a$ gives

$$\sum_{r=0}^{n-1} \frac{1}{a-r} + \sum_{r=1}^{n} \frac{1}{a+r} = H_{a+n} - H_{a-n},$$

which proves the first line. For $n > a$, only the factor $a-\nu$ in $(-\nu)_n$ vanishes at a; differentiating it and multiplying the remaining factors gives the second line, including the sign $(-1)^{a+1}$.

If $n > b = p-a-1$, the factor $(\nu+1)_n$ contains exactly one multiple of p at $\nu = a$, while all its other factors are p-units. Locally the two distinguished factors are $(-s)(p+s)$. If the product of all remaining factors is $K(s) \in \mathbb{Z}_{(p)}[s]$, then $[z^n]U = -pK(0)$ and $[z^n]V = -K(0) - pK'(0)$; hence $[z^n]V \equiv [z^n]U/p \pmod p$. In particular these coefficients of U are divisible by p, so $D_p = U \bmod p$ has degree at most b.

Finally put $n = p-1-k$, $0 \le k < a$. Wilson's theorem gives

$$\frac{(n+a)!}{p} \equiv -(a-k-1)!, \quad (n-a-1)! \equiv \frac{(-1)^{a+k}}{(a+k+1)!},$$
$$(n!)^2 \equiv \frac{1}{(k!)^2} \pmod p.$$

Combining these congruences with the factor $(-1)^{a+1}$ in the high coefficient of U yields precisely the coefficient w_k displayed above. This proves the reversed expression for W_p. □

Lemma B.3 (error decomposition)**.** *With $[\,\cdot\,]_{\ge p}$ denoting truncation to degrees $\ge p$ and $E_p = (H_p - A_p^2)/p^2 \bmod p$,*

$$E_p = \left[\tfrac{3}{8}FW_p - \tfrac{1}{16}D_p^2\right]_{\ge p} \pmod p.$$

Proof. By Theorem 4.1 and Proposition 2.4, $H_p - A_p^2 \in p^2\mathbb{Z}_{(p)}[z]$, so E_p is a well-defined polynomial in $\mathbb{F}_p[z]$, supported in degrees $\ge p$ (the coefficients of degree $< p$ of $H_p - A_p^2$ vanish exactly by Theorem 2.2). Square the expansion of Lemma B.1: the $-\frac{p}{2}FU$ term contributes $-[FW]/2$ after truncation (the low part of U does not overflow since $\deg F + \deg D = a+b = p-1$), and the $\frac{p^2}{16}(U^2 + 2FV)$ term contributes $[D^2]/16 + [FW]/8$ using $[z^n]V \equiv [z^n]U/p$

on the overflow range; E_p being the *negative* of the overflow of A_p^2 gives the displayed coefficients $3/8$ and $-1/16$. □

Lemma B.4 (reflection of the derivative part)**.** *$D_p(1-z) = (-1)^a D_p(z)$ in $\mathbb{F}_p[z]$.*

Proof. With $\mathcal{L}_a = z(1-z)\partial_z^2 + (1-2z)\partial_z + a(a+1)$, the coefficient recursion of the finite Pochhammer polynomial, together with the p-divisibility of the high coefficients of U, gives $\mathcal{L}_a D_p = -(2a+1)F$ in $\mathbb{F}_p[z]$; the last coefficient equation is consistent since $\deg D_p = b \le p-2$. Among polynomials of degree $< p$, $\ker \mathcal{L}_a$ is spanned by F (the recursion $(n+1)^2 Y_{n+1} + (a(a+1) - n(n+1))Y_n = 0$ determines Y from Y_0, all $(n+1)^2$ being nonzero). Since $\mathcal{L}_a$ commutes with $\rho : z \mapsto 1-z$ and $\rho F = (-1)^a F$ (Rodrigues), the difference $\rho D_p - (-1)^a D_p$ is both a multiple of F and anti-invariant under ρ up to the sign $(-1)^a$; in odd characteristic this forces it to vanish. □

Proof of Proposition 7.3. Apply the projection lemma to D_p^2: a polynomial $T \in \mathbb{F}_p[z]$ of degree $< 2p$ with $T(1-z) = T(z)$ writes uniquely as $L + z^p C$ with $\deg L, \deg C < p$, and comparing degrees $\ge p$ via $(1-z)^p = 1 - z^p$ gives $C(1-z) = -C(z)$. Combined with Lemma B.4 and the reversed form of W_p, the product $[FW_p]_{\ge p}$ is exactly $z^p \mathcal{Q}_a(z)$ with $\mathcal{Q}_a$ as in Section 7 (the entries of $\mathcal{Q}_a$ are the convolution of the f_i and w_k), while $[D_p^2]_{\ge p}$ has antisymmetric z^p-part. Hence

$$\widetilde{R}_p := E_p z^{-p} - \tfrac{3}{8}\mathcal{Q}_a \quad \text{satisfies} \quad \widetilde{R}_p(1-z) = -\widetilde{R}_p(z),$$

and evaluating at $z = \frac{1}{2}$ (note $(\frac{1}{2})^p = \frac{1}{2}$) gives $E_p(\frac{1}{2}) = \frac{3}{16}\mathcal{Q}_a(\frac{1}{2})$; since $H_p - A_p^2 = p^2 E_p$ in $\mathbb{Z}_{(p)}[z]$, $H_p(\frac{1}{2}) \equiv A_p(\frac{1}{2})^2 + \frac{3p^2}{16}\mathcal{Q}_a(\frac{1}{2}) \pmod{p^3}$. If a is odd, $F(\frac{1}{2}) = P_a(0) = 0$ exactly and $D_p(\frac{1}{2}) = 0$ by Lemma B.4, so $A_p(\frac{1}{2}) \equiv \frac{p^2}{16} V(\frac{1}{2}) \pmod{p^3}$ lies in $p^2\mathbb{Z}_{(p)}$ and its square is 0 modulo p^3. □

Appendix C. Proof of Theorem 7.2: the symmetric-square identity

Fix $a \ge 1$ odd, $\lambda = a(a+1)$, $s = z(1-z)$, $t = 1-2z$, and let $F = F_a(z) = \sum_{i=0}^a f_i z^i$ with $f_i = (-1)^i \binom{a}{i}\binom{a+i}{i}$, so $\mathcal{D}F = 0$ for $\mathcal{D} = s\partial_z^2 + t\partial_z + \lambda$, and let

$$\mathcal{S} = s^2\partial^3 + 3st\,\partial^2 + \big(t^2 + (4\lambda - 2)s\big)\partial + 2\lambda t$$

be the symmetric-square operator, satisfying (8) with respect to $\mathcal{D}$.

Step 1: $\mathcal{Q} := \mathcal{Q}_a$ as a positive part. Set $w_{j-1} = -1/(j^2 f_j)$ (consistent with Appendix B) and

$$V(z) = \sum_{j=1}^{a} \frac{z^{-j}}{j^2 f_j}, \qquad \text{so} \qquad \mathcal{Q} = -[FV]_+,$$

where $[\,\cdot\,]_+$ takes the nonnegative-degree part. The recursion $(j+1)^2 f_{j+1} = -(a-j)(a+j+1)f_j$ gives the exact Laurent identity

$$\mathcal{D}V = -z^{-1} + f_a^{-1} z^{-a-1}. \tag{23}$$

Step 2: forcing term. Substituting (23) into (8) with $f = F$, $v = V$: the term $-z^{-1}$ contributes $F - 3(1-z)F'$, and the nonnegative part of the $f_a^{-1} z^{-a-1}$ contribution is the constant $-(2a+1)$ (read off the top coefficients of $3sF'$, tF, $(a+1)(1-z)F$: $-3a, -2, a+1$). The operator $\mathcal{S}$ preserves the purely negative part (its possible top-degree coefficient on z^n is $(n+1)(n-2a)(n+2a+2)$, vanishing at $n=-1$ and negative beyond), so the positive-part extraction is compatible and

$$\mathcal{S}\mathcal{Q} = 3(1-z)F' - F + (2a+1). \tag{24}$$

Step 3: symmetrization. With $q = (\mathcal{Q} + \rho\mathcal{Q})/2$, $\rho Y(z) = Y(1-z)$: for odd a, $\rho F = -F$ (Rodrigues) and $\mathcal{S}\rho = -\rho\mathcal{S}$; taking the symmetric part of (24) kills the constant and gives

$$\mathcal{S}q = \tfrac{3}{2} tF' - F. \tag{25}$$

Step 4: the invariant. For any polynomial Y, the expression

$$J[Y] = 2s^2(F')^2 Y - 2s^2 FF'Y' + s^2F^2Y'' + stF^2Y' + 2\lambda sF^2Y$$

satisfies $J[Y]' = F^2\mathcal{S}Y$ identically (a formal consequence of $\mathcal{D}F = 0$: with F, F', F'' treated as independent jet variables, the difference of the two sides factors exactly as $(4sF'Y - 2sFY')(sF'' + tF' + \lambda F)$). Hence by (25),

$$(2J[q])' = 3tF^2F' - 2F^3 = (tF^3)',$$

so $2J[q] = tF^3 + C$; at $z = 0$ one has $s = 0$, $F = 1$, $t = 1$, and every term of $J[q]$ contains s, so $C = -1$:

$$2J[q] = tF^3 - 1. \tag{26}$$

Step 5: evaluation. At $z = \frac{1}{2}$: $s = \frac{1}{4}$, $t = 0$, and $F(\frac{1}{2}) = P_a(0) = 0$ (a odd), so (26) reduces to $4s^2(F')^2 q = -1$; since $q(\frac{1}{2}) = \mathcal{Q}(\frac{1}{2})$,

$$\mathcal{Q}_a\Big(\tfrac{1}{2}\Big) = -\frac{4}{F_a'(1/2)^2}.$$

Finally $F_a'(1/2) = -2P_a'(0)$. The coefficient of x in the Legendre polynomial $P_{2m+1}(x)$ is

$$[x]\,P_{2m+1}(x) = (-1)^m(2m+1)\binom{2m}{m}/4^m;$$

thus, with $a = 2m+1$,

$$\mathcal{Q}_a\Big(\tfrac{1}{2}\Big) = -\frac{16^m}{(2m+1)^2\binom{2m}{m}^2} = -\Big(\frac{2^{a-1}}{a\binom{a-1}{(a-1)/2}}\Big)^2. \quad \square$$

Appendix D. Computational checks

All symbolic claims in this paper are proved in full generality in the text. In addition, the following were verified by exact machine computation (Python, standard-library rational arithmetic, from the defining convolution (1); independent PARI/GP reimplementations agree): the coefficients of Theorem 2.2 up to degree 24; the identity (11) for $r \le 24$; the center bridge

and the defect formula (12) for the 24 odd primes $p \leq 97$; the transfer, reflection, forcing equation, kernel and integral invariant of Sections 4–5 for $3 \leq p \leq 31$; the values in Appendix A; and all nine formulas of the conjecture for the primes $5 \leq p \leq 43$ including the quadratic-form reconstructions. These checks are diagnostic; no general statement in the paper rests on them.

Selected algebraic components of the argument (the operator pullback (9) after clearing denominators, the dyadic recursion behind (11), and the divisibility assemblies of Sections 6–7) have been checked in Lean; the proof as a whole has not been formalized.

References

1. Gert Almkvist and Wadim Zudilin, *Differential equations, mirror maps and zeta values*, Mirror symmetry. V, AMS/IP Stud. Adv. Math., vol. 38, Amer. Math. Soc., Providence, RI, 2006, pp. 481–515.
2. A. Babei, L. Roy, H. Swisher, B. Tobin, and F. Tu, *Supercongruences arising from Ramanujan–Sato series*, arXiv preprint (2025), arXiv:2408.08844v3.
3. Heng Huat Chan, Yoshio Tanigawa, Yifan Yang, and Wadim Zudilin, *New analogues of Clausen's identities arising from the theory of modular forms*, Adv. Math. **228** (2011), 1294–1314.
4. Sarah Chisholm, Alyson Deines, Ling Long, Gabriele Nebe, and Holly Swisher, *p-adic analogues of Ramanujan type formulas for* $1/\pi$, Mathematics **1** (2013), 9–30.
5. Jesús Guillera, *A method for proving Ramanujan series for* $1/\pi$, arXiv preprint (2018), arXiv:1807.07394v4.
6. Guo-Shuai Mao, *Proof of some conjectural congruences involving binomial coefficients and Apéry-like numbers*, Front. Math. **21** (2026), 341–368.
7. Guo-Shuai Mao and Jia-Yan Liu, *Congruences involving Apéry-like numbers of two kinds*, 2025, Preprint, DOI 10.13140/RG.2.2.17934.42565/1.
8. Guo-Shuai Mao and Jia-Jia Yang, *Supercongruences involving Apéry-like sequences* $\{m^n G_n(x)\}$ *and* $\{m^n V_n(x)\}$, Period. Math. Hungar. **87** (2023), 303–314.
9. Robert Osburn and Wadim Zudilin, *On the (K.2) supercongruence of Van Hamme*, J. Math. Anal. Appl. **433** (2016), 706–711, arXiv:1504.01976v2.
10. Fernando Rodriguez-Villegas, *Hypergeometric families of Calabi–Yau manifolds*, Calabi–Yau varieties and mirror symmetry, Fields Inst. Commun., vol. 38, Amer. Math. Soc., Providence, RI, 2003, pp. 223–231.
11. Zhi-Hong Sun, *Congruences concerning Legendre polynomials II*, arXiv preprint (2012), arXiv:1012.3898v2.
12. ———, *New congruences involving Apéry-like numbers*, arXiv preprint (2020), arXiv:2004.07172v2.
13. ———, *Congruences concerning binomial coefficients and binary quadratic forms*, arXiv preprint (2022), arXiv:2210.17255v4.
14. ———, *Generalizations of the Christoffel–Darboux formula and congruences involving Apéry-like numbers*, arXiv preprint (2026), arXiv:2608.13192v1.
15. Zhi-Hong Sun and Dongxi Ye, *Supercongruences via Beukers' method*, arXiv preprint (2024), arXiv:2408.09776v4.
16. The LMFDB Collaboration, *The L-functions and modular forms database, elliptic curve label* `49.a1` *and cm data*, https://www.lmfdb.org/EllipticCurve/Q/49/a/1, 2026, Accessed 2026.

17. Lucien Van Hamme, *Some conjectures concerning partial sums of generalized hypergeometric series*, p-adic functional analysis, Lecture Notes in Pure and Appl. Math., vol. 192, Dekker, New York, 1997, pp. 223–236.

School of Information & Network Engineering, Anhui Science and Technology University, Fengyang, Anhui 233100, People's Republic of China
Email address: zhenghm@ahstu.edu.cn